12TH BIOLOGY · CHAPTER 16 · FULL CHAPTER MCQS

Human Endocrine System

67 topic-wise MCQs from the complete chapter with answers and reasons

16.1 HORMONES – THE CHEMICAL MESSENGERS

1. Coordination carried out by hormones is called:

(a) Nervous coordination
(b) Chemical coordination
(c) Reflex action
(d) Saltatory conduction

Answer: (b) Chemical coordination — The endocrine system sends chemical instructions.

2. A hormone shows its effects only in its:

(a) Target cells
(b) Blood cells
(c) Gland cells
(d) Nerve cells

Answer: (a) Target cells — Only target cells have its specific receptors.

3. Glands that release their secretions directly into blood are:

(a) Exocrine glands
(b) Salivary glands
(c) Endocrine glands
(d) Sweat glands

Answer: (c) Endocrine glands — They are ductless.

4. Hormones were discovered in 1902 by:

(a) Watson and Crick
(b) Schleiden and Schwann
(c) Banting and Best
(d) Bayliss and Starling

Answer: (d) Bayliss and Starling — They discovered secretin.

5. The first hormone discovered was:

(a) Insulin
(b) Thyroxine
(c) Secretin
(d) Gastrin

Answer: (c) Secretin — It stimulates the pancreas to secrete pancreatic juice.

6. Peptide and protein hormones are:

(a) Water-soluble
(b) Lipid-soluble
(c) Insoluble
(d) Derived from cholesterol

Answer: (a) Water-soluble — They dissolve easily in blood.

7. Which is a peptide/protein hormone?

(a) Cortisol
(b) Testosterone
(c) Thyroxine
(d) Insulin

Answer: (d) Insulin — Insulin is made of chains of amino acids.

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8. Glycoprotein hormones consist of a protein combined with a:

(a) Lipid (b) Carbohydrate
(c) Nucleic acid (d) Steroid

Answer: (b) Carbohydrate — They are conjugated proteins.

9. Which is a glycoprotein hormone?

(a) TSH (b) Insulin
(c) Epinephrine (d) Aldosterone

Answer: (a) TSH — TSH, FSH and LH are glycoproteins.

10. Amino acid derivative hormones are usually derived from:

(a) Glycine or alanine (b) Cholesterol
(c) Tyrosine or tryptophan (d) Glucose

Answer: (c) Tyrosine or tryptophan — They come from a single amino acid.

11. Which amino acid derivative can cross the cell membrane?

(a) Epinephrine (b) Norepinephrine
(c) None of these (d) Thyroxine

Answer: (d) Thyroxine — Thyroxine and tri-iodothyronine can cross.

12. Steroid hormones are derived from:

(a) Amino acids (b) Cholesterol
(c) Glucose (d) Nucleotides

Answer: (b) Cholesterol — Cholesterol is a type of lipid.

13. Which hormone is a steroid?

(a) Glucagon (b) Oxytocin
(c) Cortisol (d) Growth hormone

Answer: (c) Cortisol — Cortisol comes from the adrenal cortex.

14. Steroid hormones give instructions directly to the:

(a) Nucleus (b) Cell membrane
(c) Ribosome (d) Blood

Answer: (a) Nucleus — They are lipid-soluble and pass through the membrane.

15. Compared with neurotransmitters, hormones are often:

(a) Faster acting with shorter effects (b) Not chemical messengers
(c) Faster acting with longer effects (d) Slower acting with longer effects

Answer: (d) Slower acting with longer effects — Both are chemical messengers.

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16.2 ENDOCRINE SYSTEM OF MAN

16. The master gland is controlled by the:

(a) Thyroid (b) Hypothalamus
(c) Adrenal cortex (d) Cerebellum

Answer: (b) Hypothalamus — The hypothalamus is a part of the brain.

17. Cells that conduct nerve impulses and also secrete hormones are:

(a) Neurosecretory cells (b) Beta cells
(c) Interstitial cells (d) Osteoclasts

Answer: (a) Neurosecretory cells — They are present in the hypothalamus.

18. Hypothalamic hormones reach the anterior pituitary through:

(a) Axons (b) Ducts
(c) Blood vessels (d) Lymph

Answer: (c) Blood vessels — Axons carry hormones to the posterior pituitary.

19. Somatostatin inhibits the secretion of:

(a) Prolactin (b) FSH
(c) TSH (d) Growth hormone

Answer: (d) Growth hormone — It is an inhibiting hormone of the hypothalamus.

20. Prolactin inhibiting factor is actually:

(a) Serotonin (b) GABA
(c) Dopamine (d) Glycine

Answer: (c) Dopamine — It inhibits the secretion of prolactin.

21. TRH causes the secretion of:

(a) TSH (b) ACTH
(c) LH (d) GH

Answer: (a) TSH — TRH is thyrotropin releasing hormone.

22. The stalk that attaches the pituitary to the hypothalamus is the:

(a) Corpus callosum (b) Hilum
(c) Pons (d) Infundibulum

Answer: (d) Infundibulum — The pituitary has three lobes.

23. Growth hormone is also called:

(a) Thyrotropin (b) Somatotropin
(c) Corticotropin (d) Vasopressin

Answer: (b) Somatotropin — It stimulates the growth of muscles and bones.

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24. ACTH stimulates the:

(a) Thyroid gland (b) Ovaries
(c) Adrenal medulla (d) Adrenal cortex

Answer: (d) Adrenal cortex — It causes the secretion of corticosteroids.

25. FSH and LH are together called:

(a) Gonadotropins (b) Corticosteroids
(c) Androgens (d) Catecholamines

Answer: (a) Gonadotropins — They act on the gonads.

26. The hormone that stimulates mammary glands to produce milk is:

(a) Oxytocin (b) Estrogen
(c) Prolactin (d) LH

Answer: (c) Prolactin — Oxytocin causes milk ejection.

27. ADH is also called:

(a) Somatotropin (b) Vasopressin
(c) Thyrotropin (d) Calcitriol

Answer: (b) Vasopressin — It stimulates water retention by the kidneys.

28. The posterior pituitary:

(a) Synthesizes ADH
(b) Secretes TSH
(c) Secretes MSH
(d) Only stores and releases hormones of hypothalamus

Answer: (d) Only stores and releases hormones of hypothalamus — It does not synthesize any hormone.

29. MSH is produced by the:

(a) Anterior pituitary (b) Posterior pituitary
(c) Intermediate pituitary (d) Pineal gland

Answer: (c) Intermediate pituitary — It stimulates melanin production.

30. Over-secretion of GH in a growing child causes:

(a) Gigantism (b) Dwarfism
(c) Acromegaly (d) Cretinism

Answer: (a) Gigantism — Height is well above average.

31. Over-secretion of GH in adult life causes:

(a) Gigantism (b) Acromegaly
(c) Dwarfism (d) Goitre

Answer: (b) Acromegaly — Bones grow in thickness.

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32. Under-secretion of ADH causes:

(a) Diabetes mellitus (b) Addison's disease
(c) Diabetes insipidus (d) Tetany

Answer: (c) Diabetes insipidus — Excessive dilute urine is produced.

33. The butterfly-shaped gland in front of the neck is the:

(a) Thymus (b) Thyroid
(c) Parathyroid (d) Pineal

Answer: (b) Thyroid — It lies on either side of the trachea.

34. The more active and powerful thyroid hormone is:

(a) T_4 (b) TSH
(c) Calcitonin (d) T_3

Answer: (d) T_3 — T_4 is converted into T_3.

35. Calcitonin lowers the blood level of:

(a) Calcium (b) Glucose
(c) Sodium (d) Potassium

Answer: (a) Calcium — It stimulates the absorption of calcium into bones.

36. Hypothyroidism during foetal and childhood development causes:

(a) Myxoedema (b) Cretinism
(c) Goitre (d) Acromegaly

Answer: (b) Cretinism — Stunted growth and mental retardation.

37. Goitre results from the deficiency of:

(a) Calcium (b) Iron
(c) Iodine (d) Sodium

Answer: (c) Iodine — It is the swelling of the thyroid gland.

38. Exophthalmia may result from:

(a) Cretinism (b) Addison's disease
(c) Cushing disease (d) Graves' disease

Answer: (d) Graves' disease — It is the protrusion of the eyeballs.

39. The number of parathyroid glands is:

(a) Four (b) Two
(c) Six (d) One

Answer: (a) Four — They lie on the posterior surface of the thyroid.

40. Parathormone stimulates which cells to release calcium from bone?

(a) T cells (b) Beta cells
(c) Melanocytes (d) Osteoclasts

Answer: (d) Osteoclasts — They dissolve the calcium crystals of bone matrix.

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41. Under-secretion of parathormone can lead to:

(a) Tetany (b) Goitre
(c) Gigantism (d) Cushing disease

Answer: (a) Tetany — Muscles remain in a contracted state.

42. The endocrine patches of the pancreas are called:

(a) Graafian follicles (b) Islets of Langerhans
(c) Renal pyramids (d) Seminiferous tubules

Answer: (b) Islets of Langerhans — They contain alpha and beta cells.

43. Insulin is secreted by:

(a) Alpha cells (b) Interstitial cells
(c) Beta cells (d) Exocrine cells

Answer: (c) Beta cells — Alpha cells secrete glucagon.

44. Conversion of glucose to glycogen is called:

(a) Glycogenolysis (b) Gluconeogenesis
(c) Glycolysis (d) Glycogenesis

Answer: (d) Glycogenesis — Insulin promotes it in the liver.

45. Glucagon mainly promotes the:

(a) Hydrolysis of stored glycogen in liver
(b) Synthesis of glycogen
(c) Uptake of glucose by cells
(d) Conversion of glucose to fats

Answer: (a) Hydrolysis of stored glycogen in liver — This releases glucose into the blood.

46. Over-secretion of insulin results in:

(a) Hyperglycaemia (b) Hypoglycaemia
(c) Hypercalcemia (d) Diabetes insipidus

Answer: (b) Hypoglycaemia — Blood glucose falls below normal.

47. In Type II diabetes, the body cells have:

(a) No insulin receptors at all (b) No beta cells
(c) Reduced sensitivity to insulin (d) Excess glucagon

Answer: (c) Reduced sensitivity to insulin — Insulin levels in blood are normal.

48. Epinephrine and norepinephrine are secreted by the:

(a) Adrenal cortex (b) Adrenal medulla
(c) Thyroid (d) Pancreas

Answer: (b) Adrenal medulla — They prepare the body for fight or flight.

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49. Hormones of the adrenal cortex are called:

(a) Gonadotropins (b) Prostaglandins
(c) Androgens (d) Corticosteroids

Answer: (d) Corticosteroids — They include glucocorticoids and mineralocorticoids.

50. Aldosterone stimulates the kidneys to reabsorb:

(a) Potassium (b) Calcium
(c) Sodium (d) Glucose

Answer: (c) Sodium — Water follows, so blood volume is maintained.

51. Over-secretion of cortisol causes:

(a) Cushing disease (b) Addison's disease
(c) Graves' disease (d) Myxoedema

Answer: (a) Cushing disease — It occurs when the pituitary makes too much ACTH.

52. Under-secretion of cortisol leads to:

(a) Cushing disease (b) Tetany
(c) Acromegaly (d) Addison's disease

Answer: (d) Addison's disease — There is muscle weakness and loss of salts.

53. Estrogen is secreted by the:

(a) Corpus luteum (b) Placenta only
(c) Graafian follicle (d) Interstitial cells

Answer: (c) Graafian follicle — It is secreted in response to FSH.

54. Progesterone is produced by the:

(a) Graafian follicle (b) Corpus luteum
(c) Seminiferous tubules (d) Thymus

Answer: (b) Corpus luteum — It is also produced by the placenta during pregnancy.

55. Testosterone is secreted by the:

(a) Interstitial cells (b) Seminiferous tubules
(c) Corpus luteum (d) Adrenal medulla

Answer: (a) Interstitial cells — They lie among the seminiferous tubules.

56. Anabolic steroids are synthetic versions of:

(a) Estrogen (b) Thyroxine
(c) Cortisol (d) Testosterone

Answer: (d) Testosterone — They increase muscle mass, strength and endurance.

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57. Thymosin stimulates the maturation of:

(a) B cells (b) T cells
(c) RBCs (d) Platelets

Answer: (b) T cells — It is secreted by the thymus gland.

58. Melatonin is secreted by the:

(a) Pituitary gland (b) Thymus gland
(c) Pineal gland (d) Thyroid gland

Answer: (c) Pineal gland — It regulates sleep patterns.

59. Which hormone stimulates the gallbladder to release bile?

(a) Cholecystokinin (b) Secretin
(c) Gastrin (d) Renin

Answer: (a) Cholecystokinin — It is secreted by the small intestine wall.

60. Erythropoietin stimulates the production of:

(a) Bile (b) WBCs
(c) Insulin (d) RBCs

Answer: (d) RBCs — It is produced by the kidneys.

61. Hormone-like substances produced by almost all cells of the body are:

(a) Endorphins (b) Androgens
(c) Prostaglandins (d) Corticosteroids

Answer: (c) Prostaglandins — They are locally-acting vasodilators.

16.3 FEEDBACK MECHANISM IN ENDOCRINE SYSTEM

62. In positive feedback, the response:

(a) Stops the controlling system
(b) Speeds up the action of the controlling system
(c) Has no effect
(d) Destroys the hormone

Answer: (b) Speeds up the action of the controlling system — Example: oxytocin during childbirth.

63. An example of positive feedback is the release of:

(a) Oxytocin (b) Glucagon
(c) Insulin (d) Calcitonin

Answer: (a) Oxytocin — Stronger contractions cause more oxytocin release.

64. The positive feedback of oxytocin continues until:

(a) Blood glucose is normal (b) Puberty
(c) Ovulation (d) The birth of the baby

Answer: (d) The birth of the baby — Then the stimulus ends.

65. In negative feedback, the end product:

(a) Speeds up the process (b) Is destroyed
(c) Stops the initial step of the process (d) Is stored

Answer: (c) Stops the initial step of the process — It is common in the endocrine system.

66. When blood glucose goes above normal, the islets release:

(a) Glucagon (b) Epinephrine
(c) Cortisol (d) Insulin

Answer: (d) Insulin — Beta cells sense the rise.

67. Normal blood glucose level makes the alpha cells:

(a) Release more glucagon (b) Release insulin
(c) Stop releasing glucagon (d) Divide

Answer: (c) Stop releasing glucagon — This is negative feedback.

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